

Lab 3

Relational & Logical operators and conditional

Learning Objectives

The following types will be covered in this lab session

- To understand **relational operators** in C++.
- To learn **logical operators** and their use in decision making.
- To implement **conditional statements (if, if-else, nested if)**.
- To solve problems using conditions in programming.

Relational Operators

Relational operators are used to **compare two values**. The result of a relational operation is either **true (1)** or **false (0)**. These operators are mainly used in decision-making and loops.

They help the program decide conditions like whether one value is greater, smaller, or equal to another.

Syntax:

```
a == b // equal to
a != b // not equal to
a > b // greater than
a < b // less than
a >= b // greater than or equal to
a <= b // less than or equal to
```

Example:

```
#include <iostream>
using namespace std;
int main() {
    int a = 10, b = 5; // Declare variables
    cout << (a > b) << endl; // 1 (true)
    cout << (a < b) << endl; // 0 (false)
    cout << (a == b) << endl; // 0 (false)
    return 0; }
```

2. Logical Operators

logical operators are used to **combine multiple conditions**. They are used when we need to check more than one condition at a time in decision-making.

They return either true or false based on combined conditions.

Syntax:

&& // AND (both conditions must be true)

|| // OR (any one condition true)

! // NOT (reverses condition)

Example:

```
#include <iostream>
using namespace std;
int main() {
    int a = 10, b = 5;
    cout << (a > 5 && b < 10) << endl; // 1 (true)
    cout << (a < 5 || b < 10) << endl; // 1 (true)
    cout << !(a > b) << endl;          // 0 (false)
    return 0;
}
```

3. Conditional Statements (if, if-else, nested if)

if Statement:

The **if statement** is used to execute a block of code only when a condition is true. If the condition is false, the block is skipped

Syntax:

```
if (condition) {  
    // code executes if condition is true  
}
```

Example:

```
#include <iostream>  
using namespace std;  
int main() {  
    int num = 10;  
    if (num > 0) {  
        cout << "Number is positive" << endl;  
    }  
    return 0;  
}
```

If-else- Statement:

The **if-else statement** is used when we have **two choices**. If the condition is true, one block executes; otherwise, the else block executes.

Syntax:

```
if (condition) {  
    // true block  
} else {  
    // false block  
}
```

Example:

```
#include <iostream>  
using namespace std;  
int main() {  
    int num;  
    cout << "Enter a number: ";  
    cin >> num;  
    if (num % 2 == 0) {  
        cout << "Even number" << endl;  
    } else {  
        cout << "Odd number" << endl;  
    }  
    return 0;  
}
```


Nested If Statement:

A **nested if** is an if statement inside another if statement. It is used when multiple conditions depend on each other.

Syntax:

```
if (condition1) {  
  
    if (condition2) {  
        // code  
    }  
}
```

Example:

```
#include <iostream>  
using namespace std;  
  
int main() {  
  
    int age = 20;  
  
    if (age >= 18) {    //check if the condition true  
  
        if (age <= 60) { //check if the 2nd condition true  
            cout << "Adult" << endl;  
        }  
    }  
    return 0;}
```

In this lab, we studied relational operators, logical operators, and conditional statements in C++. We learned how relational operators compare values, how logical operators combine conditions, and how conditional statements help in decision-making. These concepts are essential for building logical and real-world programs.

Lab 3: Tasks

Task 1: Comparison / Relational Operators

```
#include <iostream>           // Header file for input/output
using namespace std;         // Standard namespace
int main(){                  // Main function
    int num1, num2;          // Variables for two numbers
    cout << "Enter first number: "; // Ask user
    cin >> num1;              // Input first number
    cout << "Enter second number: "; // Ask user
    cin >> num2;              // Input second number
    if(num1 == num2)          // Compare equal
        cout << "First number is equal to second number" : // Output

    if(num1 > num2)           // Compare greater
        cout << "First number is greater than second number" << endl;
    if(num1 < num2)           // Compare less
        cout << "First number is less than second number" << endl;
    return 0;                // End program
}
```

Task 2: Logical Operators

```
#include <iostream>           // Header file

using namespace std;         // Standard namespace

int main()                   // Main function
{
    int a, b;                // Variables

    cout << "Enter first number: "; // Ask user
    cin >> a;                 // Input

    cout << "Enter second number: "; // Ask user
    cin >> b;                 // Input

    if(a > 0 && b > 0)        // AND operator

        cout << "Both numbers are positive" << endl; // Output

    if(a > 0 || b > 0)        // OR operator

        cout << "At least one number is positive" << endl;

    if(!(a == b))             // NOT operator

        cout << "Numbers are not equal" << endl;

    return 0;                // End
```


Task 3: If Statement

```
#include <iostream>           // Header file

using namespace std;         // Standard namespace

int main()                   // Main function
{
    int num;                 // Variable

    cout << "Enter a number: "; // Ask user

    cin >> num;              // Input

    if(num > 0)              // Check condition
    {
        cout << "Number is positive" << endl; // Output
    }

    return 0;               // End
}
```

Task 4: If-Else Statement

```
#include <iostream>           // Header file

using namespace std;         // Standard namespace

int main()                   // Main function
{
    int num;                 // Variable

    cout << "Enter a number: "; // Ask user
}
```

```
cin >> num;                // Input

if(num % 2 == 0)            // Check even

    cout << "Number is even" << endl; // Output
else                          // Otherwise

    cout << "Number is odd" << endl;

return 0;                    // End
}
```

Task 5: Else-If Statement

```
#include <iostream>          // Header file

using namespace std;         // Standard namespace

int main(){                  // Main function

    int marks;               // Variable

    cout << "Enter marks: "; // Ask user

    cin >> marks;             // Input

    if(marks >= 80)           // Condition 1

        cout << "Grade A" << endl; // Output

    else if(marks >= 60)      // Condition 2
```

```

        cout << "Grade B" << endl;    // Output

    else if(marks >= 40)                // Condition 3

        cout << "Grade C" << endl;    // Output

    else                                // Otherwise

        cout << "Fail" << endl;        // Output

    return 0;                           // End

}

```

Task 6: Nested If Statement

```

#include <iostream>                    // Header file

using namespace std;                  // Standard namespace

int main()                            // Main function
{

    int age;                          // Variable for age

    char nationality;                  // Variable for nationality

    cout << "Enter age: ";            // Ask user

    cin >> age;                        // Input age

    cout << "Enter nationality (P for Pakistani): "; // Ask user

```

```
cin >> nationality;
```

```
// Input character
```

```
if(age >= 18) {
```

```
// Outer if
```

```
    if(nationality == 'P')
```

```
// Inner if
```

```
        cout << "Eligible for CNIC" << endl; // Output
```

```
    else
```

```
// Inner else
```

```
        cout << "Not Pakistani" << endl;
```

```
}
```

```
else
```

```
// Outer else
```

```
    cout << "Under age" << endl;
```

```
    return 0;
```

```
// Ended Successfully
```

```
}
```